

Part 2: The Perceptron Arena and Permutation Sabotage

TIMING		
Write-up reading, warm-up	05 min	
2.1 Epoch 1 Ledger	10 min	
2.2 Dynamic Accuracy Ledger	10 min	
2.3 Competitive Event	20 min	

The Classical Perceptron Framework

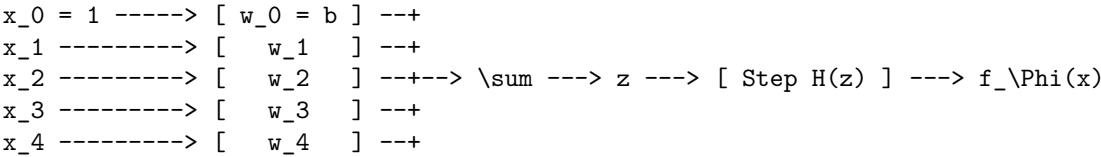
The Rosenblatt Perceptron processes augmented input vectors $\mathbf{x} = [1, x_1, \dots, x_l]^\top$ using weights $\mathbf{w} = [b, w_1, \dots, w_l]^\top$:

$$f_\Phi(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{w}^\top \mathbf{x} > 0 \\ 0 & \text{otherwise} \end{cases}$$

The learning rule processes training pairs $(\mathbf{x}^k, a^k)$ sequentially:

$$\mathbf{w} \leftarrow \mathbf{w} + (a^k - f_\Phi(\mathbf{x}^k))\mathbf{x}^k$$

THE PERCEPTRON UNIT



You will work with the same calibration dataset $\mathcal{D}_2 = \{(\mathbf{x}^k, a^k)\}_{k=1}^6$:

- $\mathbf{x}^1 = [255, 128, 128, 0]^\top, \quad a^1 = 0$
- $\mathbf{x}^2 = [55, 128, 128, 128]^\top, \quad a^2 = 0$
- $\mathbf{x}^3 = [192, 128, 128, 0]^\top, \quad a^3 = 0$
- $\mathbf{x}^4 = [100, 128, 128, 100]^\top, \quad a^4 = 1$
- $\mathbf{x}^5 = [30, 64, 128, 30]^\top, \quad a^5 = 1$
- $\mathbf{x}^6 = [20, 64, 128, 0]^\top, \quad a^6 = 1$

2.1 The Official Epoch 1-2 Ledger Challenge Initialize $\mathbf{w}^{(0)} = [0, 0, 0, 0]^\top$ and $b^{(0)} = 0$. Execute Epochs 1 and 2 sequentially on normalized samples $k = 1, 2, 3, 4, 5, 6$.

For each step k : 1. Compute the activation $z^k = b + \mathbf{w}^\top \tilde{\mathbf{x}}^k$. 2. Determine prediction $\hat{a}^k = f_\Phi(\tilde{\mathbf{x}}^k)$. 3. Calculate error $e^k = a^k - \hat{a}^k$. 4. Determine parameter updates Δb^k and $\Delta \mathbf{w}^k$. 5. Record updated parameters $b^{(k)}$ and $\mathbf{w}^{(k)}$.

Complete the **Master Epoch Ledger**:

epoch 1

Step k	Sample $\tilde{\mathbf{x}}^k$	Target a^k	Activation z^k	Output $\hat{a}^k$	Error e^k	Δb	$\Delta \mathbf{w}^\top$	New Bias b	New Weights $\mathbf{w}^\top$
1	$\tilde{\mathbf{x}}^1$	0							
2	$\tilde{\mathbf{x}}^2$	0							
3	$\tilde{\mathbf{x}}^3$	0							
4	$\tilde{\mathbf{x}}^4$	1							
5	$\tilde{\mathbf{x}}^5$	1							
6	$\tilde{\mathbf{x}}^6$	1							

epoch 2

Step k	Sample $\tilde{\mathbf{x}}^k$	Target a^k	Activation z^k	Output $\hat{a}^k$	Error e^k	Δb	$\Delta \mathbf{w}^\top$	New Bias b	New Weights $\mathbf{w}^\top$
1	$\tilde{\mathbf{x}}^1$	0							
2	$\tilde{\mathbf{x}}^2$	0							
3	$\tilde{\mathbf{x}}^3$	0							
4	$\tilde{\mathbf{x}}^4$	1							
5	$\tilde{\mathbf{x}}^5$	1							
6	$\tilde{\mathbf{x}}^6$	1							

2.2 Dynamic Accuracy Ledger After *each sample step* $k \in \{1, 2, 3, 4, 5, 6\}$ from epochs 1 and 2, compute the overall accuracy of the current parameters across the entire dataset $\mathcal{D}$. Plot or graph the accuracy curve over step number. Why does accuracy fluctuate rather than increasing smoothly?